

AntiCopyPaster

Terminal UI design for a Java refactoring extension

Quick presentation: what we are creating, why it matters, who it is for, and why the new design works.

What are we creating?

- A Java extension that reacts when a developer pastes code.
- It checks if the pasted code is duplicated somewhere else.
- It suggests an Extract Method refactoring to remove the repetition.
- Before applying it, the user can read the suggestion, inspect the diff, and choose what to do.

Why does it matter?

- Repeated code is easy to create when we copy and paste.
- Later, if one copy changes and the other does not, bugs can appear.
- Refactoring helps keep the code cleaner and easier to maintain.
- But automatic refactoring can feel risky, so the UI must build trust.

Who are we creating it for?

- Java developers using an IDE or terminal-based workflow.
- Students and junior developers who may not notice duplicated code immediately.
- Busy developers who want a quick explanation, not a long report.
- Careful developers who want proof before accepting a code change.

Main audience need: “Tell me what changed, why it is safe, and let me decide.”

Design goals from the audience

Fast to understand

The first screen shows the problem, location, and suggestion.

Control

The user can apply, edit, inspect the full diff, reject, or ask for help.

Trustworthy

Validation results show that the suggestion was checked.

Not overwhelming

Details are hidden until the user expands them.

New terminal design

Duplicate code detected after paste

You pasted code that is similar to an existing block in this file.

DETECTION

Type-2 clone

Existing: UserValidator.java:42-58
Pasted: UserValidator.java:101-117

VALIDATION

✓ Useful Extract Method

✓ Compiles

✓ Tests passed

SUGGESTION

Extract common logic into:

private boolean validateInput(...)

PREVIEW

- duplicated validation logic

+ return validateInput(input, pattern);

► Details: clone evidence, confidence, agent checks

[A] Apply [E] Edit [D] Full diff [R] Reject [?] Help

Why this design works

1. It matches the moment

2. It builds trust

3. It supports different users

4. It keeps the user in control